

Effects of Hf and C on microstructure and mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ refractory medium-entropy alloys sintered by spark plasma sintering

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ARTICLE INFO

Keywords:

Spark plasma sintering
Refractory medium-entropy alloy
Composite
Microstructures
Mechanical properties
Elevated temperature

ABSTRACT

The synergistic optimization of high temperature strength and room temperature plasticity is the core challenge of refractory high-entropy alloys (RHEAs) in practical engineering applications. In this study, four kinds of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ refractory medium-entropy alloys (RMEAs) were prepared by powder metallurgy process. The formation mechanism of oxides and carbides in refractory alloys prepared by powder metallurgy was studied. The effects of Hf and C contents on the microstructure evolution and mechanical properties of the alloys were investigated. The results show that the microstructure of the prepared $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ alloys is mainly composed of BCC matrix, carbides and dispersed nano-sized oxides. The carbides can be classified into two types: M_2C and MC. The introduction of Hf element in the form of HfH_2 can avoid the increase of oxygen during the sintering process and form dispersed hafnium oxide, which significantly improves the strength and high-temperature softening resistance of the alloy. Hf content has a key influence on the microstructure of carbides. Higher Hf content can induce the formation of MC carbides and improve the high temperature mechanical properties of the alloy. Due to the strengthening of carbides and dispersed oxides, the four $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ alloys exhibit good high temperature strength and outstanding high temperature softening resistance at 1450 °C. The yield strength at 1450 °C is more than 400 MPa, and no strain softening occurs. The compressive fracture strain of the alloy at room temperature is more than 28 %, which has good formability and damage tolerance. The prepared refractory alloy exhibits good high-temperature mechanical properties and room-temperature plasticity, and has application potential in the field of ultra-high temperature structural materials.

1. Introduction

The development of aerospace and nuclear energy industries has put forward more stringent requirements for high-temperature materials. For example, the operating temperature of hypersonic aircraft nose and scramjet engines is close to 2000 °C. However, the limit temperature of nickel-based superalloys currently used is only 1200 °C [1–4]. Although refractory alloys, ultra-high-temperature ceramics (UHTCs), and carbon-carbon (C-C) composites can withstand temperatures exceeding

1400 °C, these materials still fall short of fully meeting practical application requirements. For instance, refractory alloys such as TaW and IrRe exhibit moderate plasticity and relatively high damage tolerance at room temperature. However, their mechanical properties degrade rapidly under elevated temperatures, rendering them inadequate for high-temperature strength requirements. Ultra-high-temperature ceramics possess the requisite strength for extreme thermal environments. However, these materials exhibit poor room-temperature plasticity, limited damage tolerance, and significant fabrication challenges.

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However, carbon-carbon composites suffer from inadequate oxidation resistance at ultra-high temperatures, and their surface protection remains a critical challenge demanding urgent resolution. Consequently, the development of high-strength, high-toughness metallic materials capable of operating within the 1400 °C–2000 °C range carries significant engineering value and scientific importance.

Refractory medium/high-entropy alloys (RM/HEAs) demonstrate exceptional high-temperature strength, positioning them as promising candidate materials for ultra-high-temperature applications. In 2011, Senkov et al. [5] investigated the refractory high-entropy alloys NbMoTaW and VNbMoTaW. It is found that the compressive strength of NbMoTaW refractory high-entropy alloy can reach 400 MPa at 1600 °C, and the compressive strength of NbMoTaWV refractory alloy at 1600 °C is 479 MPa, which is significantly higher than that of traditional refractory alloys. However, the fracture strain of NbMoTaW and NbMoTaWV refractory high-entropy alloys at room temperature is only about 2.1 %, which seriously limits their processing capacity and practical application range. To enhance the room-temperature ductility of NbMoTaW RHEAs, studies have shown that the introduction of light metal elements such as Ti, Zr and Hf can effectively improve its room temperature plasticity [6–9]. At the same time, these added elements may also lead to a significant reduction in the strength of the alloy at elevated temperatures. How to improve the room temperature brittleness of refractory medium/high-entropy alloys, further enhance the high temperature strength, and solve the problem of room temperature plasticity and high temperature strength inversion is the key to the practical application of refractory medium/high-entropy alloys [10,11].

Previous studies have shown that introducing an appropriate amount of secondary ceramic carbides into ductile refractory medium/high-entropy matrices can significantly enhance the high-temperature strength of the alloy without markedly compromising its room-temperature ductility [12]. For example, the $\text{Re}_{0.5}\text{MoNbW}(\text{TaC})_{0.9}$ refractory high-entropy alloy (RHEA) designed and pretreated by Wei et al. [13] exhibited an excellent yield strength of 1.08 GPa at 1200 °C, along with a plastic strain of 8.5 % at room temperature. The $\text{MoNbW}(\text{TaC})_{0.5}$ RHEA fabricated by Luo et al. [14] demonstrated a yield strength exceeding 900 MPa at 1200 °C, with a fracture strain of approximately 10 % at room temperature. These results indicate that adding high-melting-point carbides such as TaC to TaW-based alloys with relatively low high-temperature strength can effectively improve their high-temperature mechanical properties. Building on these studies, our research group further introduced trace amounts of molybdenum (Mo): on one hand, solid solution strengthening intensifies lattice distortion to enhance creep resistance; on the other hand, the segregation of Mo at grain boundaries enables impurity purification and grain boundary strengthening, thereby mitigating the potential brittleness tendency caused by carbide addition. Meanwhile, the added trace hafnium (Hf) can form stable carbides, which not only inhibit the precipitation of harmful phases and refine grains but also improve the oxidation resistance of grain boundaries.

Based on the aforementioned multi-component synergistic design concept, we ultimately employed the arc melting method to successfully prepare a series of refractory Re-Hf-Ta-W-C alloys. These alloys demonstrate exceptional high-temperature strength while maintaining satisfactory room-temperature ductility. For example, $\text{Re}_{0.1}\text{Ta}_{1.9}\text{W}_{0.2}\text{C}_{0.4}$, $\text{Re}_{0.1}\text{Hf}_{0.1}\text{Ta}_{1.6}\text{W}_{0.4}(\text{TaC})_{0.25}$, and $\text{Re}_{0.1}\text{Hf}_{0.25}\text{Ta}_{1.6}\text{W}_{0.4}(\text{TaC})_{0.25}$ alloys exhibit exceptional mechanical performance, achieving compressive strengths of 798 MPa, 1110 MPa, and 1065 MPa at 1450 °C respectively, with corresponding room-temperature fracture strains of 17 %, 16.9 %, and 12 %. These materials exhibit good comprehensive mechanical properties and are expected to be applied to high temperature components [16–18]. However, such materials exhibit exceptionally high melting points, rendering conventional casting methods impractical for shaping them. Powder metallurgy, by contrast, offers a viable approach for near-net-shape forming of refractory components. Consequently, the successful fabrication of these materials via powder metallurgy would

hold significant scientific and technological value.

In prior research [19], our group successfully employed spark plasma sintering (SPS) to fabricate carbide-reinforced refractory medium-entropy alloys. Preliminary findings revealed that the controlled formation of oxides and carbides exerts a profound influence on the microstructure and mechanical properties of these SPS-processed alloys. In the Re-Hf-Ta-W-C refractory alloy system, hafnium serves as the critical determinant of carbide microstructure. However, when the alloy is prepared by SPS sintering, Hf is easily combined with oxygen to form oxides, resulting in a decrease in the effective Hf content of carbides, and hafnium oxide can also seriously deteriorate the microstructure and properties. Therefore, the content and relative content of Hf and C are the key to the formation of carbides and oxides in the alloy, which in turn affects the mechanical properties of the alloy.

To systematically investigate the influence of hafnium and carbon on the microstructure and mechanical properties of alloys prepared by SPS, this study employed powder metallurgy techniques to fabricate four novel refractory medium-entropy alloys with the following compositions: $\text{Re}_{0.06}\text{Hf}_{0.04}\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.01}$, $\text{Re}_{0.06}\text{Hf}_{0.1}\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.01}$, $\text{Re}_{0.06}\text{Hf}_{0.05}\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.04}$, $\text{Re}_{0.06}\text{Hf}_{0.1}\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.04}$. A comprehensive analysis was conducted to elucidate the effects of Hf and C content variations on microstructural evolution and mechanical performance.

2. Experimental procedure

2.1. Preparation of composite materials

The raw materials used included high-purity powders of Re, Ta, W, HfH_2 , and TaC. The purity and average particle sizes of the powders were as follows: Re (99.95 wt%, 1 μm), HfH_2 (99.90 wt%, 44 μm), Ta (99.95 wt%, 30 μm), W (99.95 wt%, 10 μm), and TaC (99.90 wt%, 1–3 μm). Notably, the Hf in the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs originates from the HfH_2 powder.

Herein, $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ alloys was fabricated using a powder metallurgy method (Fig. 1). The sample preparation process is based on the previous work of the author [19]. Firstly, The designed powders were weighed and ball milled using a planetary ball mill (SQMLF-4L, HuaWei, China) (Fig. 1). The ball milling process was protected by argon. The ball milling duration was 4 h, with a rotational speed of 180 rpm and a ball-to-powder weight ratio of 8:1. Furthermore, the post-milling powder handling was conducted in an argon-filled glove box.

Finally, the milled powders were sintered and densified using a spark plasma sintering (LABOX-650F, SINTERLAND, Japan) to obtain the required samples (Fig. 1). The sintering temperature of the alloy is 1800 °C, the sintering temperature holding time is 30 min, the sintering pressure is 30 MPa, and the pressure in the cooling stage is 10 MPa. Before the sintering, the mixed powders were loaded into a graphite mold with an inner diameter of 20 mm and an outer diameter of 30 mm. The graphite paper with a thickness of 1–2 mm were placed surrounding the powders to prevent the reaction and bonding of alloys powders with the graphite mold. The size of the SPS processed cylinder alloys billets was $\phi 20 \text{ mm} \times 10 \text{ mm}$.

For simplicity, the samples of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.01}$ ($x = 0.04, 0.1$), which were labeled as “CH4 and CH10”, and the samples of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_{0.04}$ ($x = 0.05, 0.1$) were labeled as “C4H5 and C4H10”, respectively.

2.2. Chemical composition and characterization of microstructure

Phase characterization was performed using X-ray diffraction (XRD, Bruker D8, Bruker Corporation, Germany) with $\text{Cu K}\alpha$ radiation. The scanning range was set from 30° to 120° at a rate of 2°/min. Quantification was conducted through the Rietveld method. The XRD specimens were rough-ground and fine-ground using a series of sandpapers (180#,

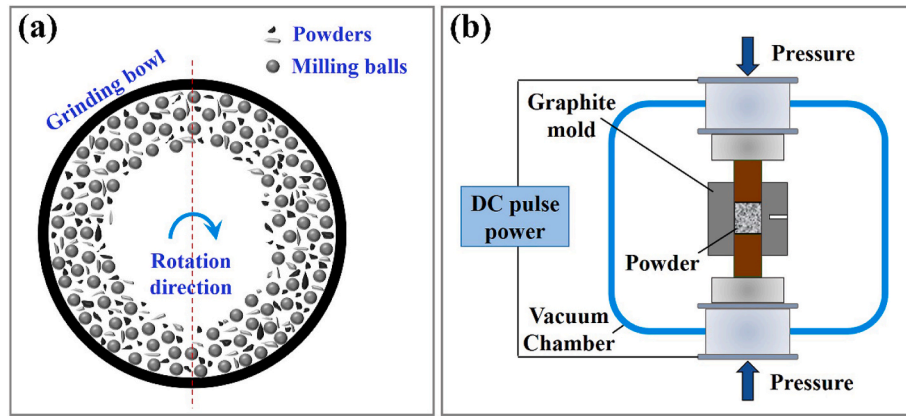


Fig. 1. Preparation process of RMEAs: (a–b) Schematic diagram of the powder-metallurgy process.

400#, 1000#, 2000#, 3000#, and 5000#), followed by mechanical polishing to eliminate surface stress and strain layers.

Microstructural and compositional characterization was conducted using scanning electron microscopy (SEM) equipped with energy-dispersive spectrometer (EDS) (SUPRA 55, ZEISS Corporation, Germany; SEM 5000, CIQTEK, China). Specimens for SEM observation were prepared through standard mechanical grinding and polishing procedures. Following polishing, samples were chemically etched at room temperature using an acidic solution (40 % H_2O + 40 % HNO_3 + 20 % HF) for approximately 10 s to reveal microstructural features.

The phase composition and microstructure of the specimens were further examined using transmission electron microscopy (TEM) (JEM-F200, JEOL, Japan). The TEM specimen preparation procedure was first sanded with sandpaper to form a thin foil with a thickness of 80 μm , and then ion thinning was performed. The parameters of ion thinning are as follows: the initial setting is 4.5 keV, the angle is 8° until the center of the specimen appears holes; then use 4 keV, angle 4° , time 20 min; finally, 3–3.5 keV was used, the angle was 3° , and the time was 30 min to obtain a better thin layer area.

Through quantitative analysis of the SEM images of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs, the average grain size of the BCC matrix, the average diameter of oxide particles, and their volume fractions in the four alloys were determined.

2.3. Characterization of mechanical properties

The density of the alloy was tested by Archimedes drainage method and the relative density of the alloy was calculated [20]. The Vickers microhardness tester (DHV-1000, Truer, China) was used to characterize the hardness of the samples. The hardness test load was 4.90 N and the test temperature was room temperature. The average value of the five tests was taken as the hardness test result.

The room-temperature compressive properties of the specimens were characterized using a universal testing machine (UTM5105X, SUNS Technology Co., Ltd, China), with WC-Co hard alloy compression anvils. And the room-temperature compression strain rate was 0.001 s^{-1} . At least three specimens were tested in each group to ensure the accuracy of the test data. The high-temperature compressive mechanical properties were characterized using a Gleeble-3500 thermal simulator, with tungsten-based ultra-high temperature anvils. The tests were conducted under argon protection, with a heating rate of $10^\circ\text{C}/\text{s}$. After reaching the target temperature, the specimens were held for 1 min to ensure uniform temperature before the compression test, at a strain rate of 0.01 s^{-1} and a test temperature of 1450°C . Both the room-temperature and high-temperature compressive specimens were cylindrical, with dimensions of $\Phi 5 \text{ mm} \times 7.5 \text{ mm}$. The surfaces of the specimens were polished using sandpapers of 180#, 400#, 1000#, and 2000# grit to ensure a smooth finish, with parallel upper and lower faces, and

perpendicularity to the side surfaces. It should be noted that fixture deformation during the compression tests may lead to strain data distortion. The data in this study has been corrected accordingly, and the specific method is detailed in previously published papers by our research group [21].

3. Experimental results

3.1. Phase analysis and microstructure

Figs. 2 and S1 present the grain size distribution histogram of the BCC matrix, the size distribution histogram of oxides in the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs, as well as the element distribution characteristics of the alloy obtained via EDS Mapping. The average grain size of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs matrix is between 2 and 3 μm , and the average size of oxide particles is about 0.5 μm . Fig. S2 confirms that the size of HfO_2 is approximately 500 nm using a large number of TEM bright-field images of HfO_2 . The SEM secondary electron images of the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs are presented in Fig. S3.

Through analysis of Figs. 2, S1 and S3, it is found that the microstructure of CH4, CH10, C4H5 and C4H10 is mainly composed of three regions with different contrast levels: the light gray regions is mainly distributed with Re, Ta and W elements, and the distribution characteristics of Re and W elements are completely consistent. The three elements of Ta, Hf and C in the dark gray area are more concentrated, and the Ta element in this area is more dense than that in the light gray area, indicating that this area may be carbide. Within the Re and W-enriched regions, Hf appears as discrete points, with sizes on the order of several hundred nanometers. These Hf-rich distribution points are completely consistent with the distribution characteristics of oxygen element, which indicates that it may be hafnium oxide (HfO_2) particles, corresponding to the black contrast part in SEM image. Notably, within the pale gray contrast regions, the corresponding EDS mapping reveals marked disparities in pixel intensity distributions for Re, Hf, and Ta, indicating that the content distribution has not yet reached the ideal uniformity. In addition, from the elemental chemical compositions of the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs, as measured using the EDS spot analyzes and listed in Table S1.

In terms of microstructure, nano-sized dispersed oxide particles are uniformly distributed in the BCC matrix, while the size of the second phase carbides reaches the micron level, and these carbides show discontinuous distribution characteristics, and there is an obvious metal transition layer between adjacent carbides.

Fig. 3 show the XRD diffraction pattern of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs. According to the results of XRD analysis, the phase composition of the four alloys is BCC matrix phase, MC/M₂C carbide phase and oxide phase. Fig. S4 shows the Rietveld refinements of observed XRD patterns. Based on the Rietveld refinements, the abundance of BCC and carbide

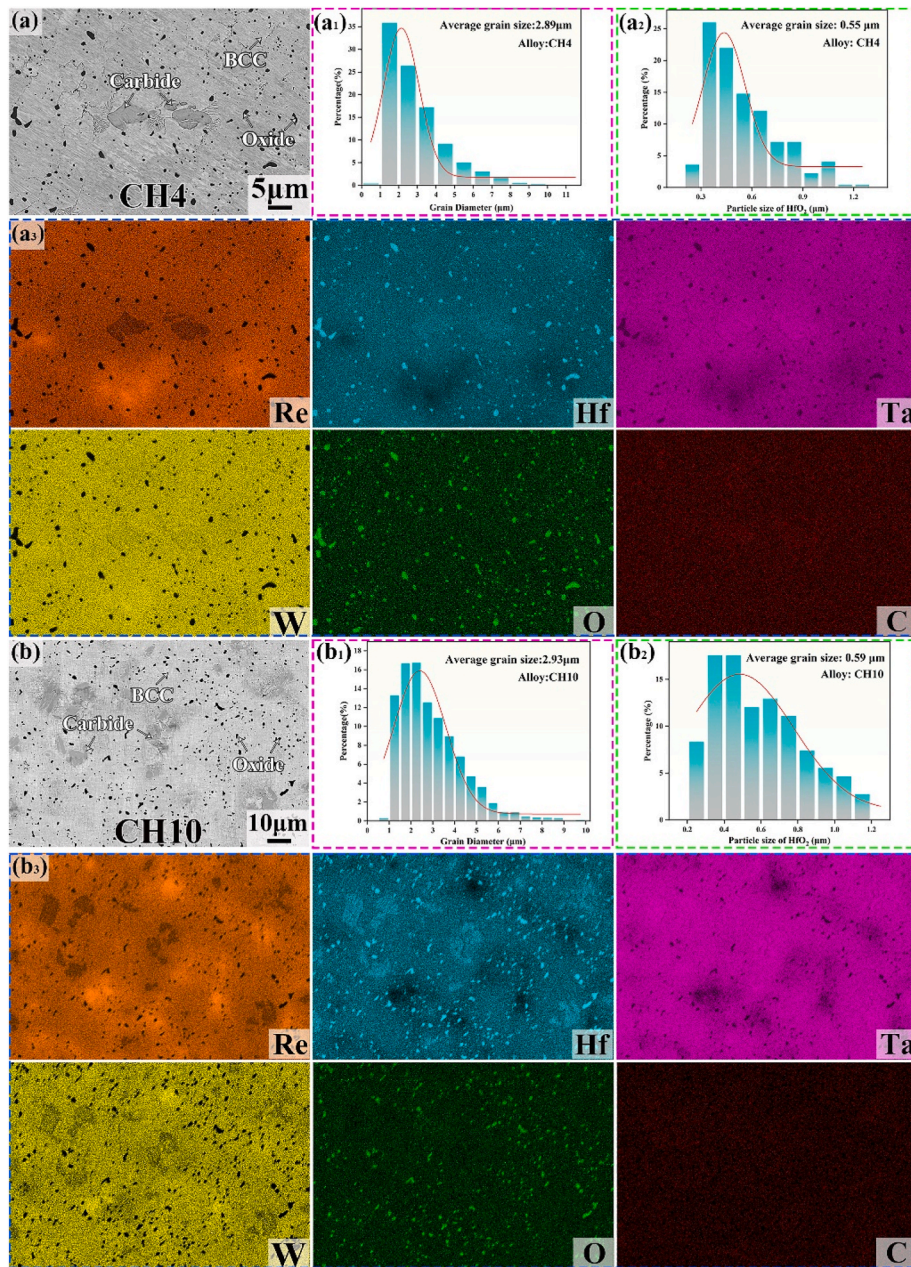


Fig. 2. (a–b) The grain size distribution histogram of the BCC matrix, the size distribution histogram of oxides in the CH4 and CH10, as well as the element distribution characteristics of the alloy obtained via EDS Mapping.

phase in the sintered $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ specimens is shown in Table S2. Among them, CH10 is mainly composed of BCC phase (JCPDS card 04–0788), FCC structure MC carbides (JCPDS card 35–0801) and oxide (JCPDS card 6–318). The CH4, C4H5 and C4H10 alloys are dominated by BCC phase, the second phase is M_2C carbides with HCP structure (JCPDS card 18–1296), and oxide phase. In addition, the XRD diffraction peak amplification diagram of Fig. 3(b) shows that the C4H10 may contain a trace amount of MC-FCC carbides. The lattice constant analysis shows that the lattice parameter of the BCC matrix is 3.26 Å, the lattice constant of MC type carbide (FCC structure) is 4.543 Å, the lattice parameters of M_2C carbides (HCP structure) are $a = 3.097$ Å and $c = 4.930$ Å.

The typical C4H10 phase structure was further characterized by TEM. The bright field images of the oxide region and the corresponding EDS data are shown in Figs. 4(a) and (b). Furthermore, the corresponding SAED pattern of the oxide and BCC phases obtained from

regions A and B are shown in Fig. 4(a). These results confirm that the oxide is HfO_2 (JCPDS card 6–318). Figs. 4(c) and (d) present the bright-field image of a carbide region in the C4H10 specimen after compression at room-temperature and its corresponding EDS mapping result, respectively. It can be seen that the interface between the carbide and the matrix is well-bonded, with no obvious defects observed. Based on the FFT image obtained at the interface in Fig. 4(e), it can be confirmed that the carbide has HCP structure and is identified as M_2C , corresponding to $(\text{Ta}, \text{Hf})_2\text{C}$. Through calculation, the BCC and HCP- M_2C phases in the composite showed the following preferred crystallographic orientation relationship along the interface.

$$(\bar{1}10)_{\text{BCC}} // (01\bar{1}0)_{\text{HCP}}, [011]_{\text{BCC}} // [2\bar{1}10]_{\text{HCP}}$$

From the IFFT pattern shown in Fig. 4(f), the interplanar spacings for the $(\bar{1}10)$ plane of BCC and the $(01\bar{1}0)$ plane of HCP- M_2C were calculated to be 0.2364 and 0.2702 nm, respectively. The corresponding

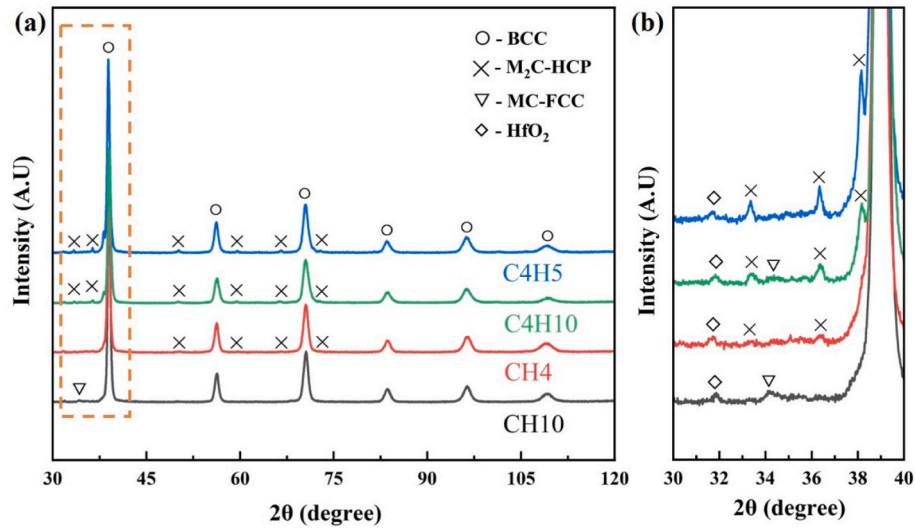


Fig. 3. XRD patterns of the sintered specimens: (a) CH₄, CH₁₀, C₄H₅ and C₄H₁₀ alloys; (b) enlargement of (a).

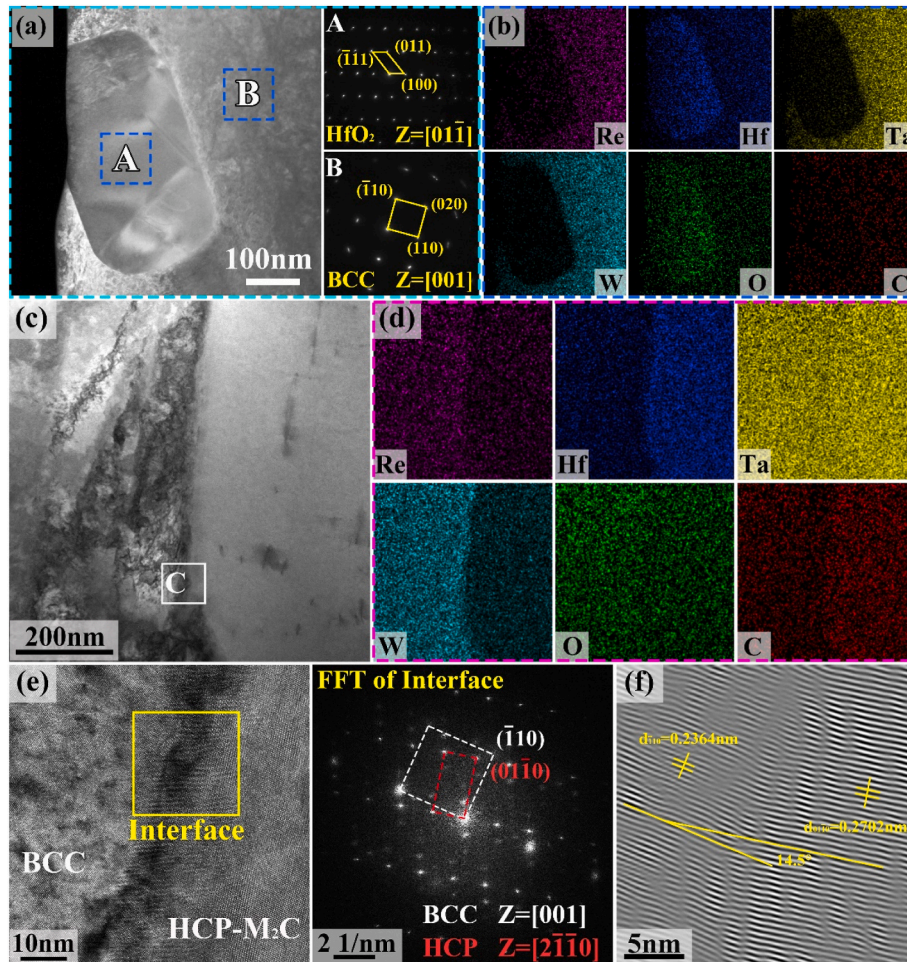


Fig. 4. (a) Bright-field image of the oxide in C₄H₁₀, and the SAED pattern of HfO₂ and BCC phases obtained from regions A and B; (b) EDS mapping data of (a); (c) TEM bright-field image of the post-test room-temperature compression specimen C₄H₁₀; (d) EDS mapping data of (c); (e) HRTEM image of the region C in (c) and FFT images of Interface; (f) IFFT pattern of HRTEM image of the Interface in (e).

orientation misfit angle was $\sim 14.5^\circ$. At the interface between the $(\bar{1}10)$ plane of BCC phase and $(01\bar{1}0)$ plane of HCP-M₂C phase, the following formula can be used to calculate the δ value of the misfit [22].

$$\delta = \frac{0.2702 - 0.2364/\cos(14.5^\circ)}{0.2364/\cos(14.5^\circ)} \approx 0.10659 \quad (1)$$

This indicates that a semi-coherent interface was formed between the

BCC and HCP-M₂C phases.

Similarly, Fig. 5 presents the TEM analysis results of another carbide region in the C4H10 specimen after room-temperature compression. It is proved that the carbide structure is FCC and is identified as MC, corresponding to (Ta, Hf)C. As shown in Fig. 5(c), the preferential crystal orientation relationship of BCC and FCC-MC phases along the interface is as follows:

$$(020)_{\text{BCC}} // (\bar{1}3\bar{1})_{\text{FCC}}, [001]_{\text{BCC}} // [013]_{\text{FCC}}$$

From the IFFT pattern shown in Fig. 5(d), the interplanar spacings for the (020) plane of BCC and the $(\bar{1}3\bar{1})$ plane of FCC-MC were calculated to be 0.1665 and 0.1388 nm, respectively. The corresponding orientation misfit angle was $\sim 5.3^\circ$. At the interface between the (020) plane of BCC phase and $(\bar{1}3\bar{1})$ plane of FCC-MC phase, the following formula can be used to calculate the δ value of the misfit [22].

$$\delta = \frac{0.1665 - 0.1388/\cos(5.3^\circ)}{0.1388/\cos(5.3^\circ)} \approx 0.1944 \quad (2)$$

This indicates that a semi-coherent interface is also formed between the BCC and FCC-MC phases. In summary, the interface between the carbides and the matrix in the alloy exhibits good bonding.

3.2. Mechanical properties

Figs. 6 and 7 show the engineering compressive stress-strain curve of the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at room-temperature and high-temperature (1450 °C), respectively. Fig. 8 shows the comparison of high-temperature strength and room-temperature ductility of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs. Fig. 9 illustrates the Vickers hardness histogram of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs. The mechanical properties of the four alloys are shown in Table 1.

As shown in Fig. 6 and Table 1, the four RMEAs prepared in this study have excellent room temperature plasticity at room-temperature. Their room temperature fracture strains exceed 28 %, especially the fracture strains of CH10 and C4H5 high-temperature alloys exceed 33 %, which is far more than most of the classical refractory high-temperature alloys. The yield strengths of CH4, CH10, C4H5 and C4H10 RMEAs at room temperature were 1157.5 MPa, 1322.3 MPa, 1224.6 MPa and 1319.4 MPa, respectively, and the compressive strengths were 2558.9

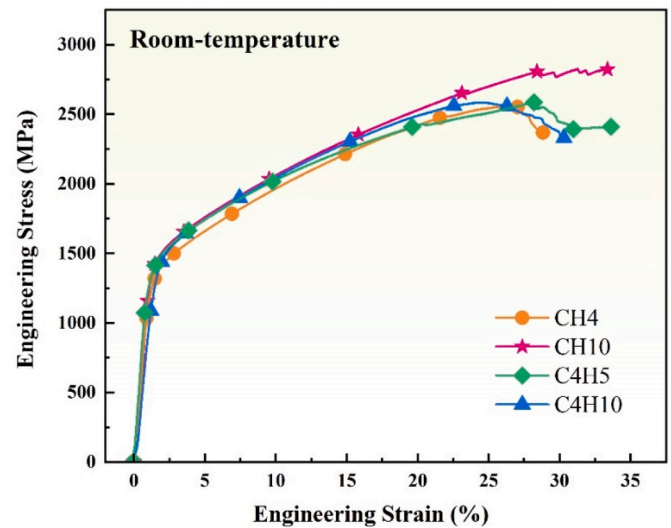


Fig. 6. Compressive stress-strain curves of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at room-temperature.

MPa, 2826.5 MPa, 2584.6 MPa and 2586.2 MPa, respectively.

Furthermore, it can be found that the yield strength of CH10 alloy is 164.8 MPa higher than that of CH4, and the ultimate compressive strength is 267.56 MPa higher than that of CH4. The yield strength of C4H10 alloy is 94.8 MPa higher than that of C4H5, and the ultimate compressive strength is similar. It is shown that the variation of Hf content has a significant effect on the yield strength and ultimate compressive strength of superalloys at room-temperature.

Fig. 7 shows the compression “engineering stress - engineering strain” curves of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at 1450 °C, and their “true stress - true strain” curves and strain-hardening curves of high temperature compression are shown in Fig. S5. CH4, CH10, C4H5 and C4H10 exhibit excellent high-temperature mechanical properties at 1450 °C. Their yield strength at high temperature exceeds 400 MPa, and the ultimate compressive strength is also above 950 MPa, which obviously shows the mechanical properties beyond the conventional

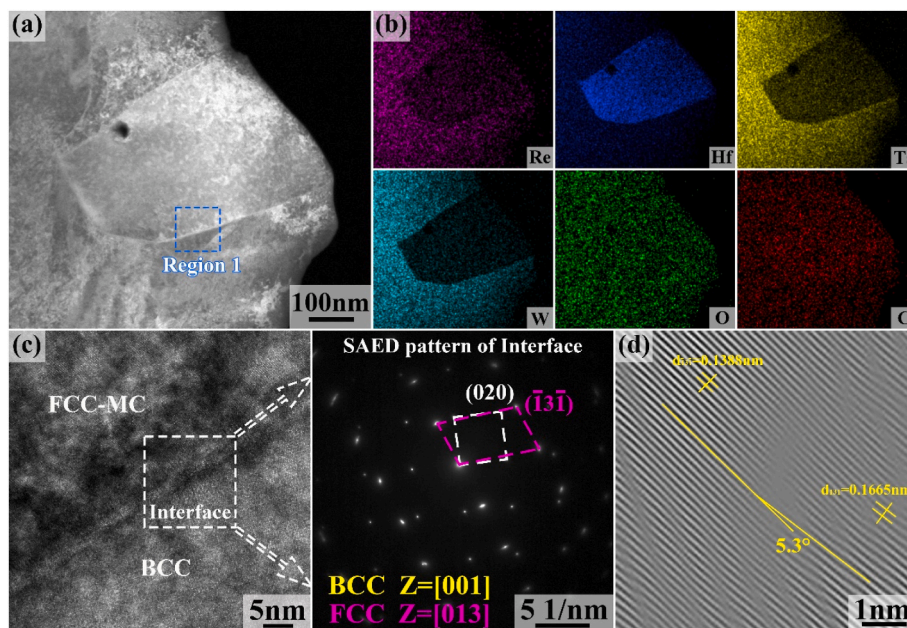


Fig. 5. (a) TEM bright-field image of the post-test room-temperature compression specimen C4H10; (b) EDS mapping data of (a); (c) HRTEM image of the region 1 in (a) and SAED pattern from the Interface; (d) IFFT pattern of HRTEM image of the Interface in (c).

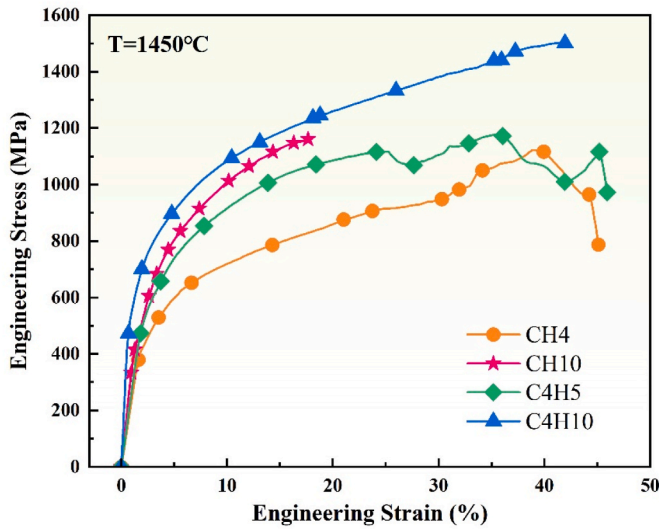


Fig. 7. Compressive “engineering stress - engineering strain” curves of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at high-temperature.

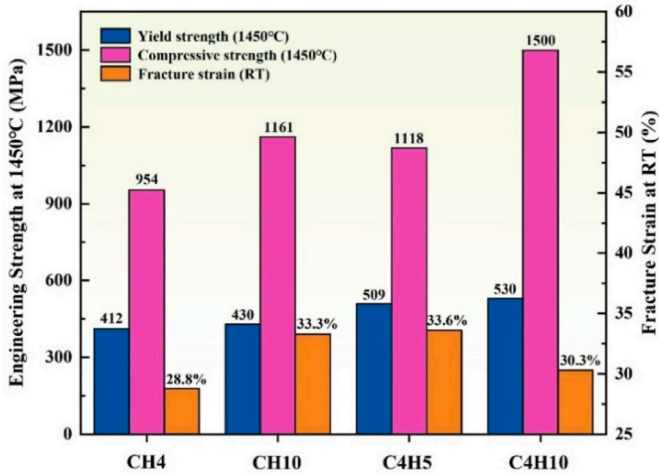


Fig. 8. Comparison of the mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at room-temperature (RT) and 1450 °C.

refractory alloys, and the four RMEAs show a certain high-temperature softening resistance. It should be noted that during the high temperature test, due to the accidental shedding of the thermocouple, the local temperature field distribution of the refractory entropy alloy specimens of CH4, CH10 and C4H5 is uneven, and the experimental process is interrupted. Although its ultimate compressive strength at 1450 °C cannot be accurately measured, its yield strength still has important reference value.

Therefore, the high-temperature strain-hardening exponent “ n ” obtained from the “true stress - true strain” curve (Fig. S5) is used to quantify the softening resistance. The “ n ” values of CH4, CH10, C4H5 and C4H10 are 0.3238 ± 0.0068 , 0.23442 ± 0.0045 , 0.3308 ± 0.0071 and 0.23499 ± 0.0052 , respectively, which is calculated by the fitting formula [23]:

$$\sigma = K \cdot \varepsilon^n \quad (3)$$

This value is higher than that of traditional materials at the same temperature ($n \sim 0.20$), indicating that it can still maintain strong strain hardening ability in high temperature plastic deformation.

From the comparative analysis of the mechanical properties in Table 1 and Fig. 8, it can be seen that the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$

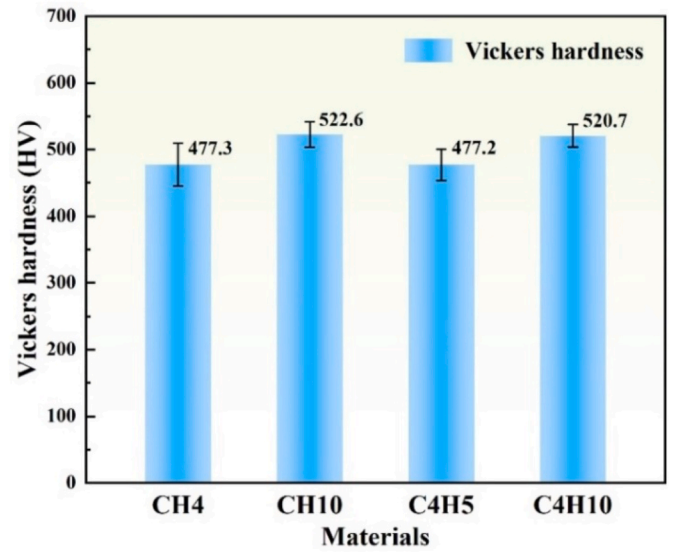


Fig. 9. Vickers micro-hardness of the sintered $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(-\text{TaC})_y$ composites.

Table 1

Mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at room-temperature (23 °C) and high-temperature (1450 °C).

RMEAs	Temperature (°C)	Yield strength (MPa)	Ultimate compressive strength (MPa)	Fracture strain (%)	Vickers Hardness (HV)
CH4	23	1157.5	2558.9	28.8	477.3
	1450	412.3	>953.8	>30.3	–
CH10	23	1322.3	2826.5	33.3	522.6
	1450	430.6	>1161.1	>17.7	–
C4H5	23	1224.6	2584.6	33.6	477.2
	1450	509.5	>1117.8	>25.24	–
C4H10	23	1319.4	2586.2	30.3	520.7
	1450	530.7	>1500.0	>40.0	–

RMEAs exhibit excellent mechanical properties at both room and high temperatures. Among the four alloys prepared, their room temperature plasticity is similar, so it is impossible to visualize the effect of MC-type carbides and M_2C -type carbides on the room-temperature plasticity of the four alloys. It can be concluded that MC-type carbide-reinforced CH10 alloy exhibit superior high temperature mechanical properties than M_2C -type carbide-reinforced CH4 alloy. In addition, the high-temperature yield strength of CH4 is lower than that of C4H5, and the high temperature yield strength of CH10 is lower than that of C4H10, indicating that the increase of C content is beneficial to enhance the high-temperature yield strength of the alloys.

As shown in Fig. 9, the hardness values of CH4 and C4H5 alloys are similar, with an average of about 477 HV, while the hardness values of CH10 and C4H10 alloys are also closer, averaging about 520 HV. Specifically, the hardness of CH10 is 45.3 HV higher than that of CH4, which is about 9.5 % higher than that of CH4. The hardness of C4H10 is 43.5 HV higher than that of C4H5, which is increased by about 9.1 %. These results indicate that the addition of Hf element has a more significant effect on the enhancement of alloy hardness in the present study system, while the change in the content of C element has a relatively small effect on the microhardness. In addition, all four alloys exhibited high densities, with relative densities in the range of 96.5 %–98.2 %. Combined with the SEM observations in Fig. S3, no defects or significant porosity are observed within the microstructure of the alloy, which further confirmed that these RMEAs had excellent microstructure integrity and compactness.

Fig. 10 shows the fracture morphology images of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs after room-temperature compression. Fig. S6 shows the specimens before and after room-temperature compression. As shown in Fig. S6, the specimen underwent very significant plastic deformation during compression at room-temperature and was finally compressed into the shape of a waist drum, and multiple irregular cracks could be observed on the cylindrical side surface of the specimen.

As can be seen from Fig. 10, the four alloys exhibit similar fracture behaviors: the carbides undergo severe tearing during compression, and the fracture surfaces show an obvious layered structure, which is similar to the typical fracture morphology of MAX phases [17]. Significant compressive deformation characteristics are observed in the carbide regions, and microcracks are detected at the layered fracture surfaces. Notably, no crack propagation is found at the interface between carbides and the matrix, indicating good bonding at the interface.

4. Discussion

4.1. Comparison of mechanical properties of the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs fabricated by SPS and other typical RHEAs at room and elevated temperatures

Fig. 11 presents a comparison of the mechanical properties between the $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs fabricated in this work and other RHEAs [5,7,15,17,18,24–26] at room temperature. As shown in the figure, the room-temperature yield strength of the four $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs ranges from 1100 to 1350 MPa, which is comparable to that of other RHEAs. However, the alloys fabricated in this work exhibit room-temperature fracture strains exceeding 28 %, significantly surpassing those of other refractory medium/high-entropy alloys. For instance, the C4H5 alloy achieves a fracture strain of 33.6 % at room temperature, representing a 1192 % increase compared to the NbMoTaW RHEA [5] and a 170.9 % enhancement relative to the NbTaW_{0.4}C_{0.25} RMEA previously developed

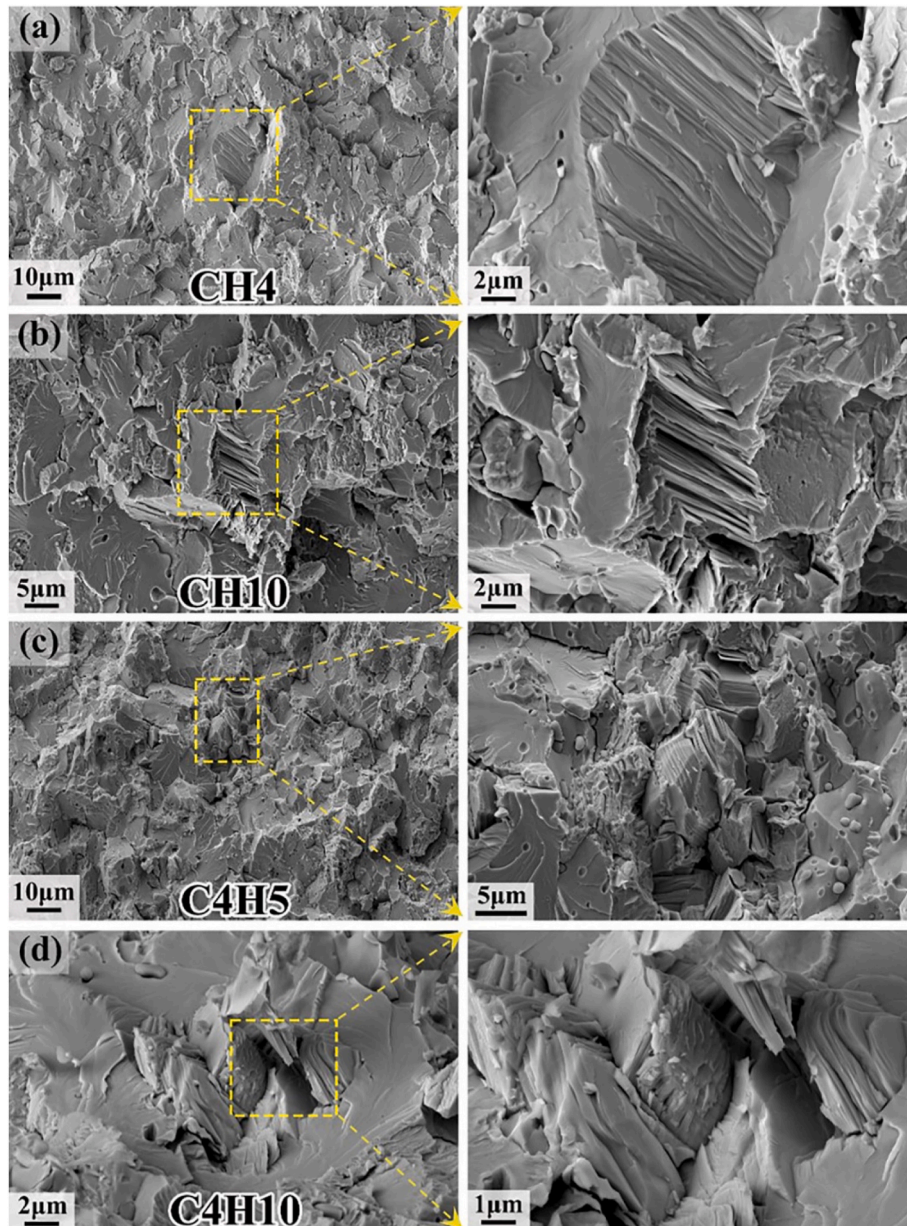


Fig. 10. Fracture morphology of post-test room-temperature compression specimens of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs.

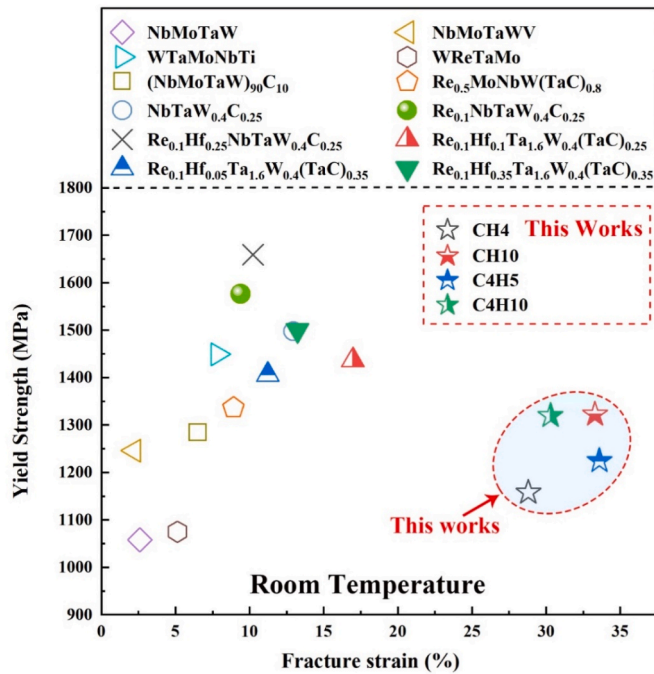


Fig. 11. The mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs were compared with other RHEAs at room temperature.

by our group [17]. The enhanced room-temperature ductility can be attributed to several factors. Primarily, the reduced tungsten concentration in the current alloy system effectively mitigates intrinsic brittleness when benchmarked against conventional NbMoTaW refractory systems. Fundamentally, NbMoTaW is a single-phase refractory alloy, which has coarse grains in the as-cast state. In addition, due to the intrinsic brittleness of the material, it is difficult to coordinate plastic deformation, and it is easy to initiate cracks and cause failure. The alloy prepared in this paper has fine and uniform grains, which is more conducive to the coordination of plastic deformation. Thirdly, NbTaW_{0.4}C_{0.25} RMEA has a high carbide content, and some carbides form a three-dimensional space skeleton, which weakens its room temperature plasticity. The alloys fabricated in this study exhibits a low carbide content with a more homogeneous dispersion, resulting in superior room-temperature ductility.

Fig. 12 shows the comparison of elevated temperature mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs and other refractory medium/high-entropy alloys [5,15–18,25,27] at 1450 °C. As shown in Fig. 12(a), at 1450 °C, the yield strength of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$

RMEAs prepared in this paper increases with the increase of carbon content and hafnium content, ranging from 400 MPa to 550 MPa. The yield strength of CH4 is the lowest, which is 412.3 MPa, which is similar to the yield strength of NbMoTaW RHEA at 1400 °C [5]. The yield strength of C4H10 is 530.7 MPa, showing outstanding high temperature strength.

Fig. 12(b) is the compressive strength of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs at 1450 °C. It can be seen that the compressive strength also increases with the increase of carbon content and hafnium content in the alloy, but the effect of hafnium is more obvious. The compressive strength of the alloy is between 900 MPa and 1500 MPa, which is significantly higher than the yield strength. Many refractory alloys exhibit limited hardening capability at elevated temperatures [28,29]. When the temperature is above 1300 °C, the yield strength of the material is often the highest point of the stress-strain curve. Beyond yielding, these materials undergo rapid softening, leading to a sharp decline in load-bearing capacity. The alloy prepared in this paper not only has good yield strength, but also has outstanding high temperature softening resistance, which is of great value for the practical application of the alloy.

The outstanding high-temperature strength of the alloy is related to the high-entropy effect formed by its multiple principal components. As indicated by the EDS analysis in Figs. 2 and S1, the alloy forms a BCC matrix through the co-solution of Re, Ta, and W elements, which induces severe lattice distortion in the crystal structure. This global distortion generates a strong stress field within the lattice, acting as a powerful barrier to dislocation movement. At high-temperatures, when second-phase strengthening may fail due to over-aging, the solid solution strengthening effect based on lattice distortion remains stable and effective, providing the alloy with long-lasting and relaxation-resistant high-temperature strength.

4.2. Oxygen migration and its effect on carbides during SPS preparation of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs

Previous studies [19] by the authors have demonstrated that oxide formation constitutes a critical factor influencing mechanical properties during spark plasma sintering of refractory alloys. Powder raw materials, ball milling process and sintering process will be accompanied by an increase in oxygen content. For $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs, the oxide modulation is more difficult due to the presence of the highly oxidizable Hf element. The author's team [19] pointed out that the introduction of Hf in the form of HfH_2 can effectively control the oxygen increase during the ball milling process. At the same time, during the sintering process, not only the oxygen increase does not occur, but also the oxygen content in the alloy is reduced. For $\text{Re}_{0.04}\text{Hf}_{0.15}\text{Ta}_{2.1}\text{W}_{0.45}(\text{TaC})_{0.45}$, the oxygen content of the powder after ball milling is only 0.40

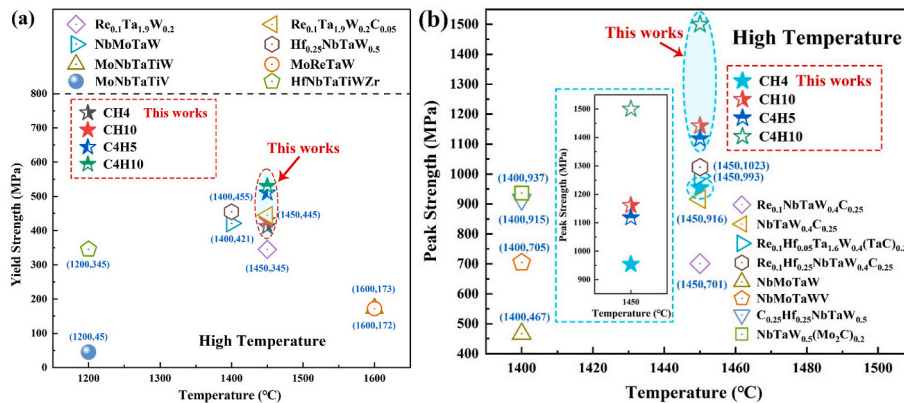


Fig. 12. Comparison of mechanical properties between $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs and other RHEAs at 1450 °C: (a) Relationship between yield strength and temperature, (b) Relationship between compressive strength and temperature.

wt%, and the oxygen content of the alloy after sintering is slightly reduced to 0.35 wt%. In addition, this method can promote the dispersion of oxide distribution and effectively reduce the oxide size.

Since the presence of oxides is difficult to completely avoid during SPS sintering, the oxygen in $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs exists in the form of HfO_2 when the Hf/O atomic ratio is higher than 0.5. At the same time, the element Hf is a key element affecting carbide formation [15], therefore, figuring out the process of oxygen migration in alloys and its effect on the organization is particularly important for regulating the organization properties of alloys.

Fig. 13 presents the curve of SPS sintering related parameters of CH4, while Fig. S7 shows the curve of SPS sintering related parameters of CH10, C4H5 and C4H10 alloys. Initially, the sample was sintered under a pressure of 10 MPa. Following vacuum evacuation, the temperature was rapidly elevated from ambient to 850 °C with a 15-min dwell time. The pressure was then increased to 30 MPa while heating to 1800 °C at a rate of 100 °C/min, followed by a 30-min isothermal hold. Then reduced the pressure and cooled to room temperature. As evident from the figure, the vacuum level began to decrease upon initiation of sintering. Activation of the vacuum pump maintained the vacuum at a relatively low level. However, after completing the 850 °C holding stage and deactivating the vacuum pump, a secondary reduction in vacuum level was observed. The initial decrease in vacuum pressure during sintering onset may result from the desorption of adsorbed gases from the powder material. A sharp decline in vacuum pressure was observed starting at 600 °C, which was due to the decomposition of HfH_2 to generate Hf and H_2 . Part of the H_2 entered the sintering chamber directly, and part of the hydrogen reduced other metal oxides except HfO_2 , and generated water vapor and discharged into the sintering chamber. The 850 °C heat preservation was to better reduce the metal oxides, and after the heat preservation, there was still gas release, which indicated that the decomposition of HfH_2 was not complete at 850 °C.

During the sintering process, the holding time at 850 °C is longer, and most of HfH_2 is decomposed into Hf and H_2 at this temperature. Hf and H_2 can also reduce other metal oxides in the raw materials. For H_2 , when escaping into the sintering chamber, it may reduce other metal oxides upon contact, producing metallic phases and H_2O . A portion of the H_2 will directly diffuse into the chamber and be eliminated by the vacuum system. In the process of H_2O escaping to the sintering cavity, if it comes into contact with Hf, it will react to generate HfO_2 and H_2 . HfO_2 is very stable, and H_2 may continue to reduce metal oxides and repeat the above process. For Hf, if there are other metal oxides, Hf will reduce metal oxides to form metal elements and HfO_2 . If there is H_2O , it will react to form HfO_2 and H_2 . If neither metal oxides nor H_2O are present, Hf will become an alloying element in the matrix or participate in the formation of carbides. The specific flow chart of oxygen migration is shown in Fig. 14. From the above discussion, it can be seen that due to the presence of HfH_2 , the oxygen content will not increase during the sintering process. However, due to the reactivity of Hf, it can react with

both metal oxides and H_2O , and the oxygen content of the sample during the sintering process is reduced to a limited extent.

The previous research of the author's team showed that Hf content is the key to affect the structure of carbides and then affect the mechanical properties. When the Hf content is low, the second phase carbides are mainly M_2C -HCP type. When the Hf content is high, the second phase carbides are mainly MC-FCC type [18]. For the alloys prepared by SPS sintering, Hf exists in the form of hafnium oxide, carbide and alloying elements under the condition of abundant Hf. Based on the preceding discussion, it can be analyzed that during the sintering process, Hf preferentially reacts with oxygen to form oxides. If all the oxygen elements in the alloy have formed HfO_2 , the remaining Hf will participate in the formation of carbides or become alloying elements. Assuming that the oxygen content in the raw material powder and the oxygen increase during ball milling are stable, the consumption of hafnium for forming hafnium oxide is constant for a sample of a certain volume or mass.

As discussed in Section 3.1, only M_2C carbides are present in both CH4 and C4H5. As can be seen from Figs. 2(a) and S1(a), although the Hf content in the carbides is higher than that in the matrix, a large amount of Hf exists in the form of HfO_2 , making it difficult to form MC-type carbides. For CH10 and C4H10 alloys, which have higher Hf contents in the alloy, as shown in Figs. 2(b) and S1(b), although a large amount of hafnium still exists in the form of HfO_2 , the Hf content in the carbides is significantly higher than that in the carbides of CH4 and C4H5 alloys. In CH10, all carbides exist in the MC-type structure. In the C4H10, due to its higher carbon content compared to CH10, both MC-type and M_2C -type carbides coexist.

The previous research by the author's team shows [18] that the microstructure of carbides has an important influence on the high-temperature mechanical properties of Re-Hf-Ta-W-C system alloys. MC-type carbides can achieve more prominent high-temperature strength, and Hf is the key to forming MC-type carbides. Therefore, when preparing $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ alloys by SPS sintering, the alloy composition needs to be specifically optimized. In particular, the content of Hf element should take into account the consumption of Hf element for forming HfO_2 . Compared with forming processes such as smelting, the SPS sintering process requires a higher content of Hf to ensure the formation of MC-type carbides.

4.3. Effects of Hf and C on the mechanical properties of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs prepared by SPS

In this paper, Hf exists in the material in the forms of oxides, carbides, and alloying elements. The author's team's previous research has confirmed [19] that the presence of large-sized HfO_2 will significantly deteriorate the mechanical properties of the alloy. In the alloys prepared in this paper, the size of HfO_2 is in the hundreds of nanometers. Compared with the alloys with similar compositions prepared by the author's team using arc melting and other methods in previous study [18], they exhibit higher room-temperature and high-temperature strength, and extremely outstanding resistance to high-temperature softening. This may be attributed to the contribution of the dispersively distributed HfO_2 . According to the discussion by Chen et al. [30], the dispersion strengthening of oxides ($\Delta\sigma_{\text{HfO}_2}$) can be explained by the Orowan mechanism, and its contribution to strength can be described by the following formula [31]:

$$\Delta\sigma_{\text{HfO}_2} = \frac{MGb}{2.36\pi} \cdot \ln\left(\frac{\varphi}{2b}\right) \cdot \frac{1}{\lambda - \varphi} \quad (4)$$

In the above formula, where M is the Taylor factor, taken as 3.06 for BCC polycrystal assemblies such as the Ta-W matrix examined here [32]. G is the shear modulus (157.7 GPa), b is the magnitude of the Burgers vector (0.276 nm), φ is the diameter of HfO_2 particles, and λ is the interparticle distance of HfO_2 particles in the Ta-W based alloys. According to the TEM and SEM analyses in Fig. S2 and S3, the HfO_2 particles can be regarded as spherical particles. For spherical particles, the

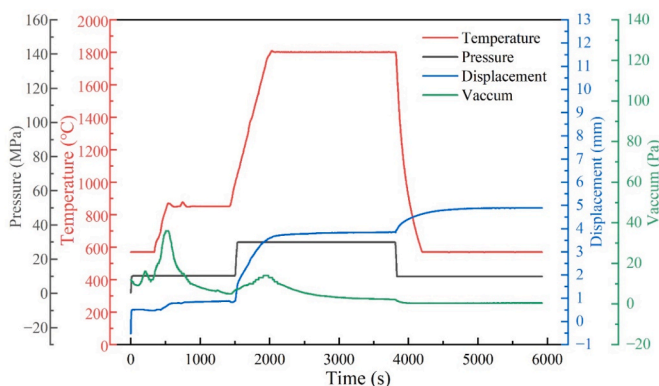


Fig. 13. SPS sintering curve of CH4.

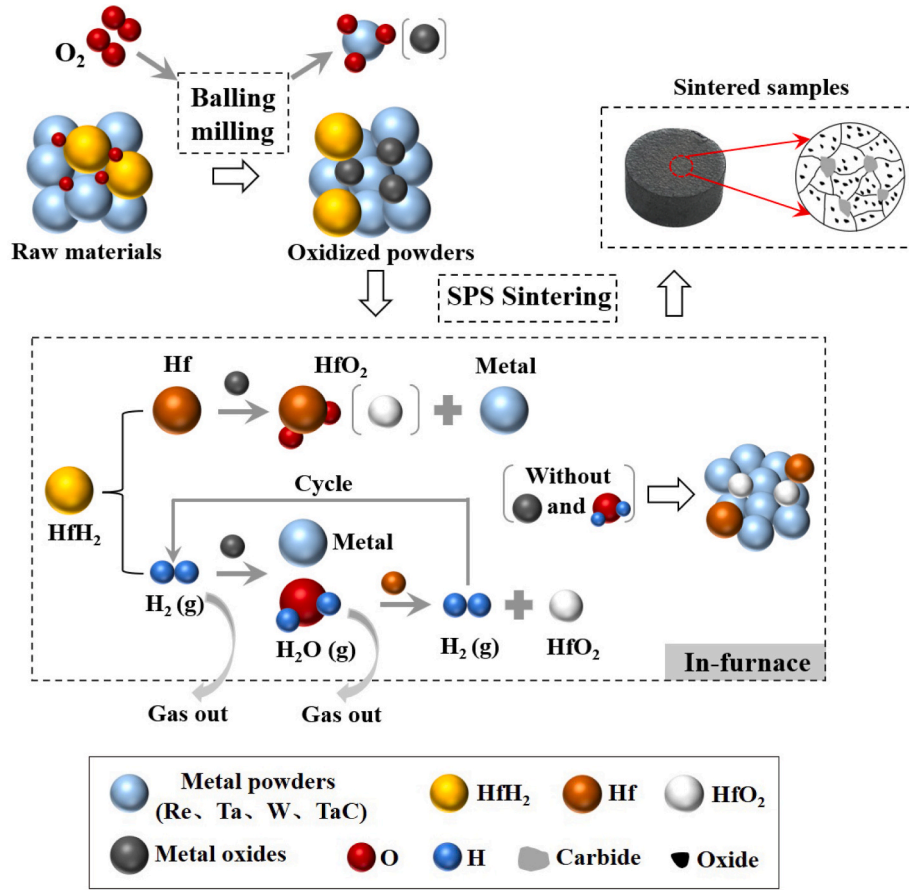


Fig. 14. Oxygen migration schematic diagram during SPS preparation of $\text{Re}_{0.06}\text{Hf}_x\text{Ta}_{1.6}\text{W}_{0.45}(\text{TaC})_y$ RMEAs.

interparticle distance can be calculated using the method in Reference [33]:

$$\lambda = \frac{1}{2} \varphi \sqrt{\frac{3\pi}{2V_{\text{HfO}_2}}} \quad (5)$$

where V_{HfO_2} is the volume fraction of HfO_2 . It should be noted that the size of HfO_2 particles in the sample is not constant. The φ value in the above formula is the arithmetic mean, which is obtained by analyzing SEM photographs. It should be noted that when determining φ and V_{HfO_2} , hafnium oxide particles larger than $1 \mu\text{m}$ were ignored to avoid underestimating the strengthening contribution of small-sized HfO_2 . Table 2 shows the parameters φ , λ , V_{HfO_2} , and the contribution value of oxide dispersion to the alloy strength $\Delta\sigma_{\text{HfO}_2}$ required for the calculations in Equations (4) and (5). As can be seen from Table 2, the contribution of hafnium oxide to the strength ranges from 35 to 45 MPa, indicating that HfO_2 makes a significant contribution to the improvement of the alloy strength. It should be noted that the contribution value of hafnium oxide to the strength in Table 2 is underestimated. This is because the size of HfO_2 particles in the sample is not constant, and the φ value in Equations (4) and (5) is obtained by analyzing SEM

Table 2
Parameters and values used in the calculations.

Parameter	Description	CH4	CH10	C4H5	C4H10
φ	HfO_2 particle diameter (μm)	0.55	0.59	0.53	0.52
λ	Interparticle distance of HfO_2 particles (μm)	3.50	4.03	3.42	3.74
V_{HfO_2}	HfO_2 volume fraction (%)	2.92	2.53	2.83	2.28
$\Delta\sigma_{\text{HfO}_2}$	Orowan strengthening by HfO_2 (MPa)	41.2	35.7	41.8	37.5

photographs as the arithmetic mean. This processing method may lead to an overestimation of the λ value.

At high temperatures, hafnium oxide exhibits exceptional stability with extremely low self-diffusion rates. Its presence not only reduces the diffusion rate of the matrix but also effectively pins dislocations to hinder their movement. Additionally, HfO_2 can pin grain boundaries to inhibit grain growth. These mechanisms collectively endow the alloy with outstanding resistance to high-temperature softening [34]. It should be specifically noted that the size of HfO_2 in this study is still relatively large, and these large-sized oxides act as sources of crack initiation (while oxides are beneficial to high-temperature mechanical strength, large-sized ones are detrimental), as illustrated in Fig. 15(a). There is no specific orientation relationship between the oxides and the matrix. Moreover, the oxides are completely brittle and hard phases, so they cannot coordinate plastic deformation. Therefore, in future research, if the size of HfO_2 can be further reduced by 1–2 orders of magnitude and its interfacial compatibility with the matrix can be improved, this will be a key direction for achieving a breakthrough in the high-temperature performance of the material.

In this paper, carbides are critical factors influencing the properties of the alloys. The introduction of carbides has no adverse effect on the plasticity of the alloy. As shown in Fig. 11, all four alloys prepared in this study exhibit better room-temperature plasticity than classical refractory alloys, and the main reasons are as follows:

Figs. 15(b) and (c) show the bright-field TEM images of the C4H10 room-temperature compression specimen and the corresponding FFT images of the corresponding regions. The obvious serrated slip trace can be observed at the edges of the carbide, which indicates that the carbide undergoes plastic deformation during the compression process. Fig. 15 (d) shows the HRTEM and IFFT images of the compressed C4H10 specimen at room-temperature. It can be seen that there are a large

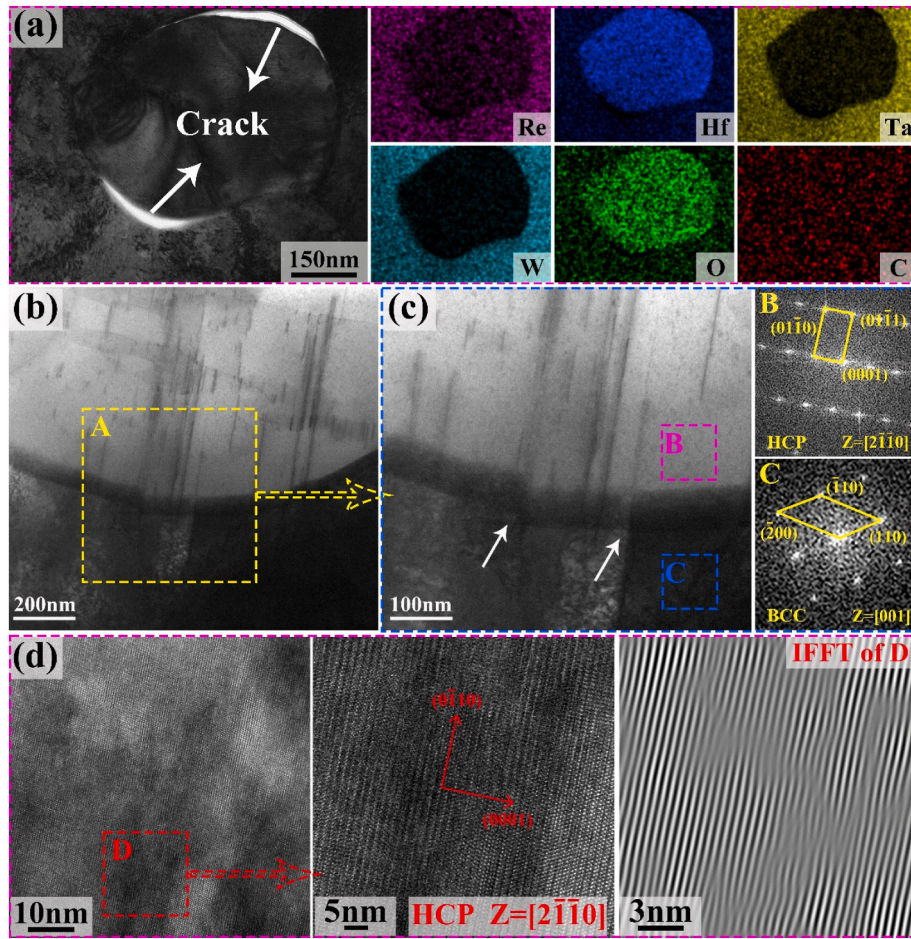


Fig. 15. (a) Bright-field images of oxides in the C4H10 specimen after room-temperature compression and the corresponding EDS mapping; (b) TEM bright field image of carbides in C4H10 after compression at room-temperature; (c) Magnification of the A and FFT images of BCC and M_2C phases obtained from regions B and C; (d) HRTEM and IFFT images of C4H10 compressed at room temperature.

number of dislocations inside the carbide, which further confirms its plastic deformation ability. In the process of macroscopic plastic deformation of the sample, carbides can coordinate with the strain of the matrix through their own deformation, thereby reducing the possibility of crack initiation.

In addition, the content and type of carbides have a significant effect on the high temperature properties of the alloy. As shown in Fig. 8 and Table 1, among the four alloys at 1450 °C, CH4 exhibits the lowest yield strength of 412.3 MPa. When the carbon content increases, the carbide content in the alloy rises, and the yield strength of the C4H5 significantly increases to 509.5 MPa. This indicates that an increase in carbide content contributes to the improvement of the alloy's high-temperature strength. The microstructural characteristics of carbides represent another critical factor influencing high-temperature mechanical properties. In both CH4 and C4H5 alloys, only M_2C carbides are present. The carbides in CH10 are MC structure, while the C4H10 contains both MC and M_2C carbides simultaneously. Although the carbon content in the CH10 alloy is comparable to that of the CH4 alloy, the carbides in CH10 exhibit an MC structure. As a result, its ultimate compressive strength at 1450 °C is 207.3 MPa higher than that of the CH4. Although the C4H5 and C4H10 alloys have comparable carbon contents, the latter exhibits a compressive strength advantage of 382.2 MPa due to the presence of MC-structured carbides.

In summary, there are two main mechanisms by which Hf affects the mechanical properties of the alloy. The first mechanism is through the formation of hafnium oxide, which in turn influences the mechanical properties of the alloy. For this mechanism, further refining the size of

hafnium oxide is beneficial to the improvement of mechanical properties. The second mechanism is through influencing the carbide structure to affect the mechanical properties of the alloy. For this mechanism, the formation of MC-type carbides is more conducive to the improvement of high-temperature strength. The influence mechanism of element C on the mechanical properties of alloys is relatively simple. The mechanism is to form carbides, thereby improving the strength of the material. However, it should be noted that the content of hafnium involved in forming carbides in the alloy has an important impact on the structure of carbides, which in turn affects the strengthening effect of carbides. When designing the composition of Re-Hf-Ta-W-C system alloys, if the spark plasma sintering preparation method is adopted, the contents of Hf and C need to be comprehensively considered.

5. Conclusion

In this study, four types of $Re_{0.06}Hf_xTa_{1.6}W_{0.45}(TaC)_y$ RMEAs were prepared by spark plasma sintering, and the effects of Hf and C on the microstructure and mechanical properties of the alloys were investigated. The main results are summarized as follows:

- (1) Introducing Hf element in the form of HfH_2 can avoid the increase of oxygen content during the sintering process and form nano-scale HfO_2 particles with dispersed distribution in the matrix. The HfO_2 particles exhibit obvious strengthening effects, contributing tens of megapascals to the strength. While effectively improving

the room-temperature and high-temperature strength, they also significantly enhance the anti-softening ability at 1450 °C.

- (2) The ratio of Hf to C significantly influences the mechanical properties of the alloy. Hf in the alloy primarily exists in the form of oxides and carbides. When the Hf content is low, oxides are preferentially formed, resulting in carbides with low Hf content, such as M_2C -type carbides. Conversely, when the Hf content is high and sufficient Hf remains after oxide formation, Hf-rich MC-type carbides can be formed. This type of carbide effectively enhances the high-temperature mechanical properties of the alloy. In addition, the interface between the carbides and the BCC matrix exhibits good bonding, and the carbides at the room-temperature compression fracture surfaces show a layered structure similar to that of MAX phases. The coordinated deformation between the plastic carbides and the BCC matrix is an important reason for the alloy to maintain excellent room-temperature plasticity.
- (3) In this paper, four types of $Re_{0.06}Hf_xTa_{1.6}W_{0.45}(TaC)_y$ RMEAs prepared by SPS exhibit excellent comprehensive mechanical properties through the strengthening of carbides and dispersed nanoscale oxides. At 1450 °C, the yield strength of all alloys is higher than 400 MPa, and they exhibit outstanding anti-softening capability with no strain softening observed. At room-temperature, they show significantly better compressive plasticity than other refractory high-entropy alloys, with a compressive fracture strain greater than 28 %. The high-temperature strength is comparable to that of NbMoTaW, but the fracture strain at room-temperature is increased by more than 10 times, demonstrating excellent room-temperature compressive plasticity. On the premise of ensuring good high-temperature mechanical properties, the excellent room-temperature compressive plasticity indicates that the material has good processing performance and damage tolerance. Such materials have the potential to manufacture high-performance high-temperature components.

CRediT authorship contribution statement

Z. Yang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **H.T. He:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **J.X. Fang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **T. Sun:** Supervision, Investigation. **B. Ma:** Validation, Investigation. **T.T. Guo:** Validation, Investigation. **W.B. Wang:** Supervision, Investigation. **T. Fu:** Visualization, Investigation. **X.Y. Zhou:** Validation, Supervision, Investigation. **G.Q. Yang:** Validation, Supervision. **J.T. Wei:** Visualization, Data curation. **M. Wen:** Visualization, Supervision, Investigation, Funding acquisition. **P. He:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work is supported by the General Program of National Natural Science Foundation of China (No. 52475193), Yunnan Fundamental Research Projects (Nos. 202201BE070001-059, 202501AT070375 and 202401BE070001-050), Yunnan Kingdian Yingcai Support Program Young Talents Special Support Project (No. KKXX202401019), Yunnan Fundamental Research Projects (No. 202301AT070342), and Scientific and Technological Project of Yunnan Precious Metals Laboratory (No. YPML-20240502017).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msea.2025.149075>.

Data availability

No data was used for the research described in the article.

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